Task 1

a

statistic	value	p_value	p_method	N_perm
Mutual Information	0.032573	0.002200	permutation	5000
Jaccard Index	0.000000	1.0	permutation	5000
Pearson χ^2	7.78365	0.005003	χ^2 parametric	*None*

MI and χ^2 both reject the null of independence at $\alpha = 0.05$ (p < 0.01).

JI $= 0$ means both columns almost never share 1's simultaneously

Conclusion: the two binary variables in p1a are dependent, but not through shared "1" events.

b

Statistic	n_significant
MI	93
JI	56
χ^2	93

Pair	count
MI & JI	55
MI & χ^2	91
JI & χ^2	56
MI & JI & χ^2	55

1. High agreement between MI and χ^2 (91 overlap of 93 χ^2 rejections):

MI and χ^2 are both sensitive to any dependence between two categorical variables, not just co-occurrence of 1's like JI.

Therefore, they detect similar dependent pairs which explains the high agreement between the two.

2. Jaccard Index much more conservative (56 pairs):

JI measures co-occurrence of 1's only.

It ignores mutual absence and other forms of dependence, so it finds fewer associations than the above two schemes.

Almost every JI-significant pair (55/56) is also MI and χ^2-significant, showing that when there is strong co-occurrence, all three metrics agree.

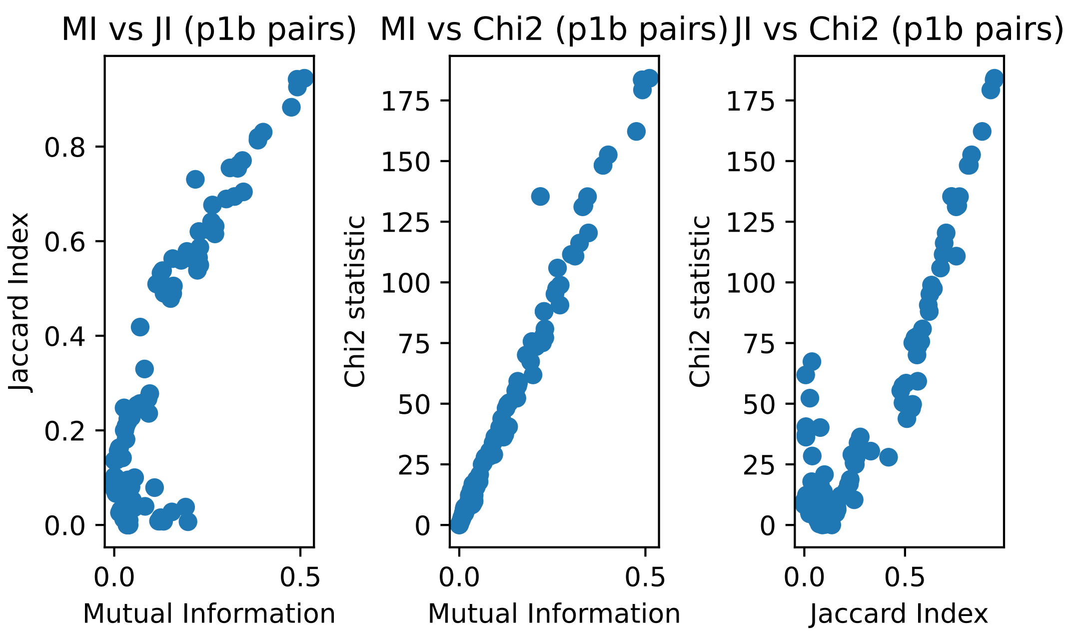
3. Patterns in overlaps:

The 55 pairs that are marked significant in all schemes correspond to genuinely strong "co-activation" relationships (both 1s appear together often).

The additional 36–38 pairs that are MI/chi^2-only likely represent inverse or asymmetric dependencies.

4. FDR correction effect:

Even after multiple-testing adjustment (105 pairs), most pairs remain to be significant, suggesting a strong dependence within the dataset.



5. Visualization analysis:

MI vs χ^2: a nearly linear positive relationship (both increase with strength of dependence).

JI vs MI/χ^2: points cluster near 0 for weak pairs, and a small subset with high JI correspond to the 55 triply significant pairs.

Task 2

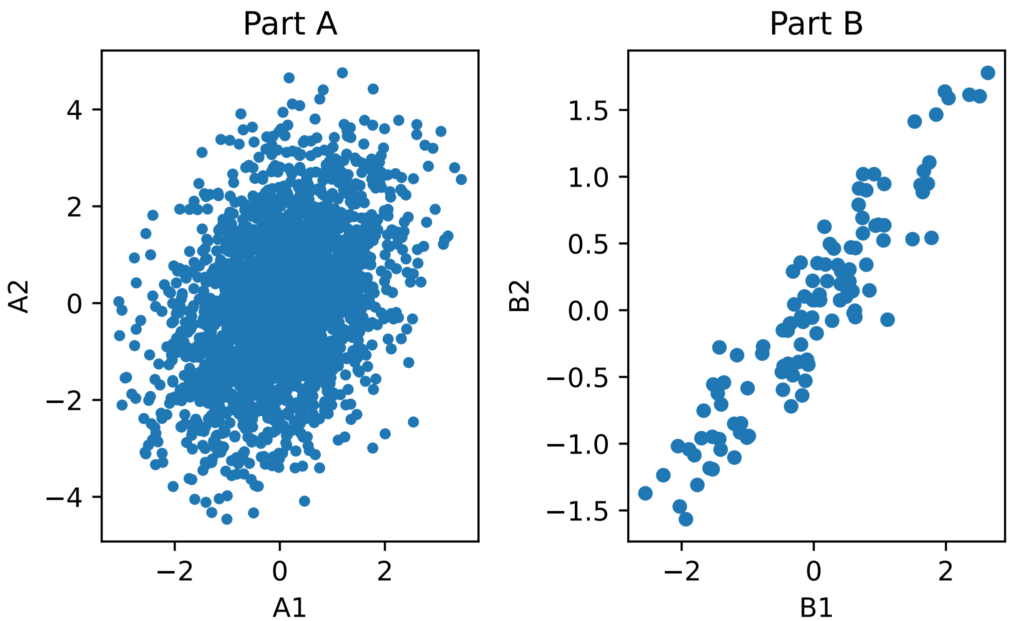
a

The pearson correlation coefficient between the two variables in p2a.csv is $r_a = 0.3809$ and the p-value was $p_a = 1.0409 \times 10^{-83}$. This means at significance level $\alpha = 0.05$ there is a statistically significant relationship between the two variables. According to the r_a coefficient, there is a weak to moderate positive association between the variables.

b

The pearson correlation coefficient between the two variables in p2b.csv is $r_b = 0.9312$ and the p-value was $p_a = 3.7373 \times 10^{-49}$. This means at significance level $\alpha = 0.05$ there is a statistically significant relationship between the two variables. According to the r_b coefficient, there is a strong positive association between the variables.

According to the correlation coefficients, the variable pair in b has a stronger association. According to the p-values, the association between the variable pair in a is less likely to have occured by chance than the association between the variables in b. The p-value is not necessarily an indicator of strength of association as p-value will decrease as n increases. As such, although the p-value and correlation coefficient disagree with eachother on which pair is more associated, this is primarily due to the fact that these two statistics measures different things. The correlation is a measure of strength of the linear relationship between two variables while the p-value is a measure of how sure we are of this measurement. Thus, the p-value is lower for a simply because $n_a > n_b$ which results in the statistic being more sure of the measurement in part a.

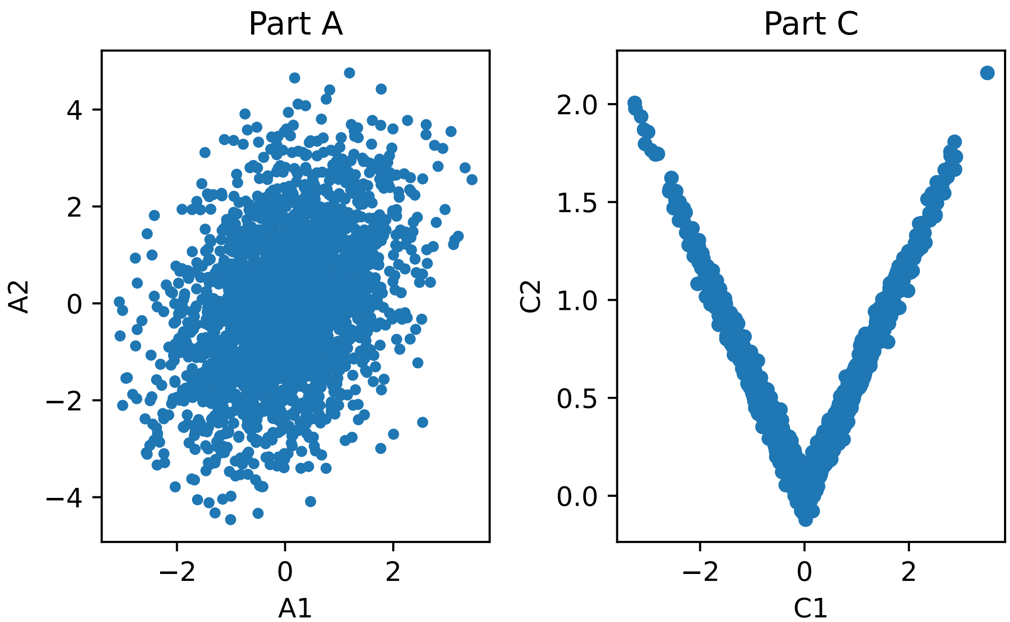


According to the scatterplots, the variables in part b have a higher association because the scatter plot is much tighter than the one for part a. This conclusion agrees with my previous statements because it is clear that the number of points in a is significantly larger than the number in b which is probably what lead to the p-value being higher in a than in b despite b having a stronger correlation than a.

c

The pearson correlation coefficient between the two variables in p2c.csv is $r_c = 0.0412$ and the p-value was $p_c = 0.5919$. This means at significance level $\alpha = 0.05$ there is not a statistically significant relationship between the two variables. According to the r_c coefficient, there is a very weak positive association / no association between the variables.

According to the correlation coefficients, the variable pair in a has a stronger association. According to the p-values, the association between the variable pair in a is less likely to have occured by chance than the association between the variables in c. The p-value is not necessarily an indicator of strength of association as p-value will decrease as n increases. For this part, the p-value and correlation coefficient agree with eachother, as they both assert that a has a higher correlation and statistical significance.



Looking at the graph, it is clear that the variables in part c have an extremely high association and roughly follow the line of $C2 = |C1|$. This is a clear disagreement to the statements made before seeing the scatterplot. Since the pearson's correlation coefficient is solving for a single straight line, the best it can do is roughly a flat horizontal line through the center of the scatter plot in c. As such, the correlation coefficient will be small (as the line of best fit has a near 0 slope), which explains the differences between the two conclusions.